

Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

Unisphere – University Club & Event Management Platform

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Unisphere is a full-stack university club and event management platform designed to centralize campus communication, simplify event organization, and improve student engagement. The platform provides a unified system where students can discover clubs based on their interests, register for events, participate in real-time group discussions, and receive instant notifications.

The system also enables faculty members and administrators to manage club activities, approve events through a structured workflow, monitor participation, and maintain institutional oversight efficiently. By replacing fragmented communication channels such as WhatsApp groups, emails, and notice boards, Unisphere creates a more organized, interactive, and scalable campus ecosystem.

The platform is developed using modern web technologies including React, Node.js, Express.js, MongoDB, and Socket.io, ensuring secure authentication, real-time communication, and responsive user experience.

In universities, information related to clubs, events, workshops, and campus activities is often shared through fragmented communication channels such as WhatsApp groups, emails, and physical notice boards. This scattered approach creates difficulties for students in accessing timely and organized information, leading to low event participation and reduced student engagement.

Additionally, new students face challenges in discovering clubs and communities that match their interests. Faculty members and administrators also struggle to manage event approvals, monitor activities, and maintain proper communication across different student organizations due to the absence of a centralized management system.

Therefore, there is a need for a unified digital platform that can streamline club management, event organization, communication, and institutional oversight within the university environment.

- To develop a centralized platform for managing university clubs, events, and student activities efficiently.
- To simplify event creation, registration, approval, and participation processes within the campus ecosystem.
- To provide real-time communication and notifications between students, clubs, faculty members, and administrators.
- To enable interest-based club discovery and improve student engagement in extracurricular activities.
- To implement a secure and scalable system with role-based access control for students, faculty, and administrators.
- To improve institutional oversight and communication by replacing fragmented systems such as emails, notice boards, and messaging groups.

- **Secure Authentication** – JWT-based login system with role-based access control for students, faculty, and administrators.
- **Club Management** – Create, manage, and explore university clubs through a centralized platform.
- **Event Management** – Create, approve, register, and manage events efficiently with structured workflows.
- **Real-Time Communication** – Live group chats, direct messaging, and instant notifications using Socket.io.
- **Student Dashboard** – Personalized dashboard for joined clubs, registered events, chats, and notifications.
- **Faculty & Admin Control** – Monitor activities, approve event requests, and manage platform operations.
- **Cloud File Uploads** – Upload event posters, profile images, and attachments using Cloudinary integration.
- **Scalable & Responsive System** – Modern responsive interface with secure and scalable full-stack architecture.

- **Frontend:** React.js with Tailwind CSS
- **Backend:** Node.js with Express.js
- **Database:** MongoDB (NoSQL Database)
- **Real-Time Communication:** Socket.io
- **Authentication:** JWT & bcrypt.js
- **Cloud Storage:** Clouinary
- **Notifications:** Firebase & Nodemailer
- **Queue & Background Jobs:** Redis with BullMQ
- **Version Control:** Git & GitHub

Development Methodology

- Agile Development Approach
- MERN Stack Architecture
- REST API Integration
- Real-Time Communication using Socket.io

Development Tools

- VS Code
- MongoDB Atlas
- Postman
- Git & GitHub

Additional Facilities

- Responsive User Interface
- Cloud-based File Storage
- Secure Authentication System
- Real-Time Notifications

- Developed a complete full-stack university club and event management platform using the MERN stack.
- Successfully implemented real-time communication and notifications using Socket.io.
- Designed a secure authentication and role-based access control system for students, faculty, and administrators.
- Built an event approval workflow to simplify institutional event management and monitoring.
- Implemented cloud-based file uploads and scalable database management using Cloudinary and MongoDB.
- Improved centralized communication and student engagement through a unified digital platform.
- Gained practical experience in frontend development, backend APIs, database design, authentication, and real-time systems.

Unisphere successfully provides a centralized and efficient platform for managing university clubs, events, and student activities. The system simplifies communication, event organization, and approval processes while improving student engagement through real-time interaction and notifications.

By integrating modern web technologies such as React, Node.js, MongoDB, and Socket.io, the platform delivers a secure, scalable, and user-friendly solution for students, faculty members, and administrators. The project also demonstrates practical implementation of full-stack development, real-time communication, authentication, and database management concepts.



Thank You